Problem 9 – ProofCouncil

Partial result: not a complete solution (unreviewed).

Rigidity of a constant boundary factor of a deformed totally geodesic surface: partial results

Abstract. Under boundary hypotheses (i)–(ii) we ask whether the constant factor Y_t of the boundary curve of a deformed totally geodesic surface $V_t \subset \mathcal{M}_g$ is independent of t. This is proved when the boundary stratum has at least two nodes or V_0 is nonarithmetic; in the remaining one-node arithmetic case it is proved only under an additional lifting property (L), so the problem is *not* fully resolved.

1 Introduction

The problem. Let $\mathcal{M}_g$ be the moduli space of genus g Riemann surfaces and $\overline{\mathcal{M}}_g$ its Deligne–Mumford compactification. The boundary $\partial\mathcal{M}_g$ is partitioned into strata Δ_G parametrizing stable curves with dual graph G; $\Delta_{G'}$ lies on the boundary of Δ_G when G' is obtained from G by pinching further curves. Throughout, closures are taken in $\overline{\mathcal{M}}_g$.

Let $V_0 \subset \mathcal{M}_g$ be a smooth complex surface, totally geodesic for the Teichmüller metric, and let V_t, t in the unit disc Δ, be a holomorphic family of subvarieties deforming it: there is a complex submanifold $\mathcal{V} \subset \mathcal{M}_g \times \Delta$ with V_t its fibre over t, and this family is assumed smoothly trivial. The V_t for $t \neq 0$ are *not* assumed totally geodesic. Two hypotheses are imposed: (i) for every boundary stratum Δ_G, the intersections $\overline{V_t} \cap \Delta_G$ form a smoothly trivial family; (ii) these boundary families are compatible with V_0: if $\Delta_{G'}$ lies on the boundary of Δ_G and $U \subset \overline{\mathcal{M}}_g$ is open with $\overline{\overline{V_0} \cap \Delta_G} \cap \Delta_{G'} \cap U = \overline{V_0} \cap \Delta_{G'} \cap U$, then for all t, $\overline{\overline{V_t} \cap \Delta_G} \cap \Delta_{G'} \cap U = \overline{V_t} \cap \Delta_{G'} \cap U$. Fix a dual graph Γ whose stratum is a product $\Delta_\Gamma = \mathcal{M}_{g_1,n_1} \times \cdots \times \mathcal{M}_{g_k,n_k}$, with projections π_j, and put $\partial V_t = \overline{V_t} \cap \Delta_\Gamma$. Suppose $\dim \partial V_t = 1$ and that for some i, $\pi_i(\partial V_t) = \{Y_t\}$ is a single point for every t. *The task is to prove $Y_t = Y_0$ for all t.*

For a boundary stratum S we write $B_S(t) = \overline{V_t} \cap S$, with its reduced structure; since V_t is algebraic, $\dim B_S(t) \leq 1$, and $\partial V_t = B_{\Delta_\Gamma}(t)$. We read "subvariety" as closed algebraic, and "smoothly trivial family" in the embedded sense: the union of the fibres is a smooth submanifold of the target times the disc, and projection to the disc is a smoothly trivial submersion (merely diffeomorphic fibres would be weaker). For the boundary families in (i), holomorphicity then *follows* (Lemma 1). All moduli spaces are coarse unless a lift to Teichmüller space is explicitly specified.

What is proved. Without further hypotheses: (1) if Γ has at least two edges, $Y_t = Y_0$ (Corollary 3); (2) if the field of definition K_N of the affine invariant manifold attached to a marked lift $N \subset \mathcal{T}_g$ of V_0 (Section 3) is not $\mathbb{Q}$, then $Y_t = Y_0$ (Corollary 11); (3) in the sole unresolved case, $\Delta_\Gamma = \mathcal{M}_{h,1} \times \mathcal{M}_{m,1}$ is a one-node separating stratum with $Y_t \in \mathcal{M}_{h,1}$, the active (moving) factor is an arithmetic Teichmüller curve in $\mathcal{M}_{m,1}$, $m \geq 2$ (one generated by a square-tiled surface), whose generating quadratic differential vanishes at the marked point (Theorem 14, Proposition 15); (4) every two-node cusp (nodal limit point) of the central ($t = 0$) one-node boundary curve is isolated in $\overline{V_0} \cap S$ for its two-node stratum S, and by (i)–(ii) has a unique holomorphic continuation $p(t)$ (Proposition 16, Corollary 17). What is *not* proved is that $p(t)$ is stationary, or that its normalization components other than Y_t are fixed. Theorem 20 proves $Y_t = Y_0$ in general under an explicit lifting property (L) on the boundary curves (Section 5); Section 7 records why (L) has not been derived from (i)–(ii).

Strategy. The mechanism is a *corner argument* (Proposition 6): if a stable curve p with more nodes lies in the closure of ∂V_0, is isolated in $\overline{V_0} \cap S$ for its own stratum S, and *stays* in $\overline{V_t} \cap S$ for small t, then Y_t lies in the finite set of smooth pointed normalization components of p. Hypothesis (ii) gives this stationarity when S has at least three nodes (Lemma 5); the hard case is a one-node stratum, whose simplest corners are two-node curves: we obtain their isolation (Proposition 16) but not their stationarity. Sections 3–4 use the boundary theory of [1], cylinder deformations [11] and a second-order sewing expansion of period matrices [4]; Section 5 introduces (L) and the rigidity of Teichmüller curve maps [6], and Section 6 runs the corner argument under (L).

2 Consequences of hypotheses (i)–(ii)

Lemma 1 (Holomorphicity of the boundary families)**.** *Under the above interpretation, the boundary families in (i) are holomorphic submersions. In particular $t \mapsto Y_t$ is holomorphic, so local constancy at 0 implies constancy on the whole disc.*

Proof. Step 1: a generic projective envelope. Embed $\overline{\mathcal{M}}_g$ in projective space and choose an affine graph $(z, F(z,t))$ for one connected component of the interior family V_t, near any chosen parameter $t = 0$. Every $F_j(\,\cdot\,, t)$ is algebraic over $\mathbb{C}(z)$. For each degree n, evaluate the finitely many monomials in (z, F_j) of degree at most n and take their Taylor-coefficient matrix in z. Linear dependence of its columns is a closed condition on t, given by holomorphic minors, and these closed sets cover the disc, so Baire's theorem and the identity theorem make all maximal minors vanish identically for some n. Over $K = \operatorname{Frac}\mathbb{C}\{t\}$, finitely many rows span the row space; a nonzero vector in their kernel annihilates every row and gives a polynomial relation for F_j. Thus $K(z, F)$ is finite over $K(z)$, and K is algebraically closed in this field: differentiating the minimal polynomial of any element algebraic over K shows that all its z-derivatives vanish, so the element is meromorphic in t alone. The prime ideal of all relations therefore defines a geometrically integral generic variety [8, Tag 037P] (reducedness is automatic in characteristic zero). Take its projective closure over $\mathbb{C}\{t\}$ and

spread its finitely many homogeneous equations over a finitely generated complex subalgebra $R \subset \mathbb{C}\{t\}$; the generic fibre is geometrically integral, since this holds after extending $\operatorname{Frac} R$ to K. Generic flatness and generic geometric integrality [8, Tags 052A, 0559, 0578] give integral fibres of dimension two after inverting some $0 \neq f \in R$. The nonzero germ $f(t)$ has no zeros on a small punctured disc, where each envelope fibre contains the original graph and equals the individual projective closure of that component by irreducibility and dimension. Repeat for the finitely many continued components; the central envelope may have extra components, none of which are excluded or used here.

Step 2: the boundary family is complex. Consequently $\bigcup_{t\neq 0} B_S(t) \times \{t\}$ is analytic. By (i) its underlying set, including $t = 0$, is an embedded smooth family. At the regular points of the analytic part its tangent spaces are complex; density and continuity of tangent spaces make them complex everywhere, including at $t = 0$, and a smooth embedded submanifold with complex tangent spaces is a complex submanifold (its local graph maps have complex-linear differentials, hence are holomorphic). Its projection, already a smooth submersion, is therefore a holomorphic submersion. This applies near any parameter. Finally, a local holomorphic section followed by π_i gives Y_t, and the identity theorem proves the last assertion. □

The next lemmas exploit the literal quantifier over open sets in (ii), applied to the particular open set $U = \overline{\mathcal{M}}_g \setminus \overline{B_S(0)}$, for which both sides of the antecedent are empty.

Lemma 2 (Fixed curves in deeper strata)**.** *If S has at least two nodes, every curve component of $B_S(t)$ is contained in $B_S(0)$. A curve component continued from $B_S(0)$ is therefore fixed, as a subset, for all small t.*

Proof. Choose a boundary stratum H obtained by smoothing one node of S, and set $U = \overline{\mathcal{M}}_g \setminus \overline{B_S(0)}$. Both sets in the antecedent of (ii), for H, S, U, are empty, since $\overline{B_H(0)} \cap S \subset B_S(0)$, so (ii) gives $B_S(t) \cap U = \overline{B_H(t)} \cap S \cap U$. The right side is finite: $B_H(t)$ has dimension at most one, and a locally closed algebraic curve has only finitely many points in its closure outside its original locus. Because $B_S(0)$ is closed in S, a curve component of $B_S(t)$ not contained in $B_S(0)$ would have an infinite, nonempty Zariski-open intersection with U, which is impossible. Such a curve component must thus be one of the finitely many curve components of $B_S(0)$, and smooth continuation cannot switch between that finite collection. □

Corollary 3. *The conclusion of the problem holds if Γ has at least two edges.*

Proof. By Lemma 2 the i-th projection of any curve component of ∂V_t is Y_0; by hypothesis it is Y_t. □

Lemma 4 (Coarse continuation in the given, unlabelled stratum)**.** *An isolated point p of $B_S(0)$ has a unique local continuous continuation in $B_S(t)$, and this continuation is holomorphic. If the curve components of $B_S(t)$ are fixed and a continuous section $p(t) \in B_S(t)$ starts on a curve component of $B_S(0)$, then $p(t) \in B_S(0)$ for all sufficiently small t.*

Proof. The smooth trivialization continues the isolated points as separate zero-dimensional components; holomorphicity (Lemma 1) makes their locally unique sections holomorphic. There are finitely many such sections. For the second assertion their central values differ from $p(0)$, so the given section stays locally in the fixed curve part. □

Lemma 5 (Stationarity beyond two nodes)**.** *If S has at least three nodes, then $B_S(t) \subseteq B_S(0)$ for all $t \in \Delta$. In particular, every isolated point of $B_S(0)$ has constant local continuation.*

Proof. Smooth one node of S to obtain a boundary stratum H with at least two nodes. Lemma 2 implies that the curve part of $B_H(t)$ is contained in $B_H(0)$; the remaining components are finitely many points of H, whose closures do not meet S. Therefore $\overline{B_H(t)} \cap S \subseteq \overline{B_H(0)} \cap S \subseteq B_S(0)$. Applying (ii) with $U = \overline{\mathcal{M}}_g \setminus \overline{B_S(0)}$, as in Lemma 2, gives $B_S(t) \cap U = \varnothing$; since $B_S(0)$ is closed in S, this proves the inclusion. A continuous section through an isolated point of $B_S(0)$ must then be constant near 0. □

Proposition 6 (A stationary corner suffices)**.** *Let S lie on the boundary of the chosen stratum Δ_Γ. Suppose $p \in \overline{B_{\Delta_\Gamma}(0)} \cap S$ is isolated in $B_S(0)$ and belongs to $B_S(t)$ for every sufficiently small t. Then $Y_t = Y_0$ for all t. In particular the conclusion of the problem holds if $\overline{B_{\Delta_\Gamma}(0)}$ contains a point with at least three nodes that is isolated in its full stratum intersection $B_S(0)$.*

Proof. Choose an open neighbourhood U of p with $B_S(0) \cap U = \{p\}$. Since p is a limit from the chosen stratum, the antecedent of (ii) holds for Δ_Γ, S, U. Thus $p \in B_S(t) \cap U = \overline{B_{\Delta_\Gamma}(t)} \cap S \cap U$. The compactified clutching map from $\prod_j \overline{\mathcal{M}}_{g_j,n_j}$ to $\overline{\mathcal{M}}_g$ is proper. For a fixed t, the image of its closed subset with i-th coordinate Y_t therefore contains $\overline{B_{\Delta_\Gamma}(t)}$. Hence the stable curve p admits a clutching presentation in which the i-th factor is the smooth pointed curve Y_t. A fixed stable curve has only finitely many such smooth pointed factors (a normalization component with a labelling of its node preimages), so Y_t takes values in a fixed finite set. Continuity fixes its value near 0, and holomorphicity (Lemma 1) extends the equality to the whole disc. For the last assertion combine Lemma 5 with the continuation in (i). □

These arguments require isolation in the *whole* stratum intersection $B_S(0)$, not just among the limits of the chosen curve; the next lemma, about a single surface, handles symmetric two-node corners without any lifted family.

Lemma 7 (Both partial smoothings at an isolated two-node point)**.** *Let $Z \subset \mathcal{M}_g$ be an algebraic surface, and let $p \in \overline{Z}$ represent a stable curve with exactly two nodes a, b. Suppose p is isolated in the intersection of $\overline{Z}$ with its stratum. In a local deformation chart with the nodes labelled, the closure of Z contains a curve approaching p on which exactly b remains nodal, and also such a curve on which exactly a remains nodal.*

Proof. A versal deformation chart of the stable curve is smooth, with plumbing coordinates u_a, u_b (one for each node, vanishing exactly when that node persists). Its map to the coarse space is finite and its boundary is $u_a u_b = 0$ [3, Sections 8–10, Proposition 12.2 and Corollary

12.3] (a chart for the single fixed surface $\overline{Z}$, not a pullback of a trivialized family). Choose a lift of p and an irreducible surface branch A of the closure of the lifted interior through that lift; such a branch exists by finiteness of the chart, and its interior is dense. Neither u_a nor u_b is identically zero on A, so the zero set of either function on A has a one-dimensional component through the chosen point, by Krull's principal ideal theorem. But $A \cap \{u_a = u_b = 0\}$ is zero-dimensional there, since the chart is finite and p is isolated in its stratum intersection. A curve component of $A \cap \{u_b = 0\}$ is consequently not contained in $\{u_a = 0\}$; its general points give the first assertion, and interchanging a, b gives the second. □

Corollary 8 (Equal-genus ends)**.** *Suppose the chosen one-node stratum is $\mathcal{M}_{h,1} \times \mathcal{M}_{h+r,1}$, with $r \geq 1$, and its whole intersection with $\overline{V_t}$ has first coordinate Y_t. Suppose it has a two-node limit, isolated in the* whole *two-node stratum intersection of $\overline{V_t}$, whose normalization is a chain $Y_t \cup_a A \cup_b W$ with $g(A) = r$, $g(W) = h$, where W, including its attaching point, is fixed. Then $Y_t = W$.*

Proof. Apply Lemma 7 to V_t, smoothing a and keeping b: this produces a curve in the same one-node stratum whose genus-h factor tends to W; that factor is identically Y_t on the entire stratum intersection, so $Y_t = W$. Since the two factors have different genera, no labels on the coarse stratum are needed. □

3 Teichmüller-geometric input on the central surface

From now on $N \subset \mathcal{T}_g$ denotes a marked lift of V_0 to Teichmüller space, a totally geodesic complex surface, and $\overline{N}$ its closure in the bordification of $\mathcal{T}_g$ used in [1]. For $X \in N$, the Teichmüller discs through X tangent to N are generated by a two-dimensional space $Q_X N$ of holomorphic quadratic differentials on X; QN is the resulting rank-two bundle and $Q_{\max} N$ the locus of differentials in QN lying in the largest (generic) stratum of quadratic differentials (by zero orders, not a boundary stratum). Arana-Herrera and Wright [1] define an equivalence relation on cylinder curves (core curves of flat cylinders) of differentials in QN: roughly, two parallel cylinders are equivalent if they remain parallel, with constant circumference ratio, on all nearby surfaces of QN, so that they can only be deformed together. We use the following consequences of their work. (1) A marked boundary intersection of $\overline{N}$ with a stratum is smooth, irreducible, algebraic, and *semisimple*: a product of Teichmüller-space factors on which the components of the pinched surface either stay fixed or move along totally geodesic submanifolds [1, Theorem 1.3]. A one-dimensional such locus has one positive-dimensional simple factor, and its *active* (nonconstant) coordinate maps are Teichmüller-disc isometries. (2) Pinched curves occur in whole equivalence classes; each class has a positive multitwist stabilizing N, and a class can be unpinched while the other nodes remain pinched [1, Corollary 2.2, Lemmas 2.6 and 2.13, Corollaries 2.12 and 3.17] (integrality of the twist exponents comes from the rationality of modulus ratios in Corollary 2.2). (3) If a nonpoint simple boundary factor is supported on subsurfaces $U_1, \ldots, U_s$, then $\partial(\bigcup_j U_j)$ is a union of whole equivalence classes [1, Lemma 6.1(3)]. Successively unpinching r distinct classes gives the *dimension bound* $\dim L \leq \dim N - r$ for a boundary locus L with

those classes pinched; see the proof of [1, Theorem 3.9]. Thus all nodes of a boundary *curve* of a totally geodesic *surface* form a single equivalence class.

Stabilizers and cusps. Let L be a marked semisimple curve, L_j an active projection, and Λ, Λ_j their setwise stabilizers in the product mapping class group and the j-th mapping class group (pinching twists already removed), divided by their pointwise kernels. The disc isometry $L \to L_j$ identifies Λ with a subgroup of Λ_j (an element acting trivially on the active disc acts trivially on L); the normalized moduli map is the orbifold covering of this inclusion, of finite degree by algebraicity, and its source has finite hyperbolic area (a torsion-free finite cover is an ordinary finite-area curve). We need a cusp only for the chosen active curve: Theorem 14 makes its differential holomorphic and its sole mark a zero; forgetting that zero, and taking the orientation double cover if necessary, changes the lattice only by finite index, so the noncompactness theorem for algebraic Teichmüller curves [10], [7, Proposition 2.4] supplies a cusp.

Lemma 9 (Forbidden leaf support)**.** *A totally geodesic surface cannot have a boundary curve in a stratum with at least two nodes whose only active normalization component corresponds to a vertex with exactly one incident half-edge and no loop.*

Proof. Let a be the unique node incident to that component Y. At a marked lift of a generic point of the proposed curve, all the nodes belong to one class by the dimension bound. On the other hand, the support assertion (3) makes $\partial Y = \{a\}$ a union of entire classes. The other node or nodes give a contradiction. □

An arithmetic reduction. Let $\mathcal{A}_N$ be the affine invariant manifold of square roots of the largest stratum of QN (an orbit closure of translation surfaces, locally linear in period coordinates), and let K_N be its field of definition in period coordinates. We use a connected component and make finite marking choices when needed. For a surface N, $\mathcal{A}_N$ has complex dimension four, rank two, and no relative tangent directions [12, Theorem 1.3 and its proof].

Proposition 10 (A singleton forces arithmeticity)**.** *If the closure of N contains a stable Riemann surface with exactly one node, then $K_N = \mathbb{Q}$.*

Proof. The node a is a singleton cylinder class. Realize it on a differential in $Q_{\max}N$. If that differential is a square, its cylinder is an entire class on the translation surface, and [11, Theorem 7.1] (the field of definition lies in the field generated by circumference ratios within a class) gives field $\mathbb{Q}$. Otherwise take the orientation cover. Its involution cannot preserve a flat cylinder: in cylinder coordinates its derivative is -1, and preservation would give a fixed regular point. Thus cylinders upstairs come in pairs and project to cylinders downstairs. The two lifts of a have equal circumferences and are equivalent. Any cylinder equivalent to them would project to an N-equivalent cylinder, since the square-root locus locally covers $Q_{\max}N$. There are no others. The same theorem, with the single ratio 1, now gives $K_N = \mathbb{Q}$. □

Corollary 11. *The conclusion of the problem holds whenever $K_N \neq \mathbb{Q}$. Consequently the only unresolved one-node case has $K_N = \mathbb{Q}$.*

Proof. A boundary point has a compatible marked lift: use a local branch of the Dehn space as in the proof of [1, Lemma 3.13]. Proposition 10 rules out a one-node stratum when $K_N \neq \mathbb{Q}$. The chosen stratum must then have at least two nodes, and Corollary 3 applies. □

4 The active differential vanishes at the attachment

Only the central totally geodesic surface is used in this section. By Corollaries 3 and 11 the remaining case is a one-node stratum; a nonseparating one-node stratum has a single factor, which cannot be constant on a curve, so the node is separating. Let $N \subset \mathcal{T}_g$ be a marked lift of V_0 with one-node boundary curve $L = \{(Y, y)\} \times D$, $D \subset \mathcal{T}_{m,1}$, where (Y, y) is the fixed pointed curve of genus $h \geq 1$, D is a Teichmüller disc, and a is the separating node; by the dimension bound the class of a is a singleton. Locally on D, use a parameter τ, write the pointed curve at τ as (E_τ, x_τ), and choose a nonzero holomorphic generator $q_D(\tau)$ of its Teichmüller differential line, a meromorphic quadratic differential on E_τ with at most a simple pole at x_τ. We recall some terminology. A quadratic differential with a double pole has a *quadratic residue* R^2, the coefficient of $(dz/z)^2$; if $R^2 \neq 0$ there is a local *exponential coordinate* w, unique up to a scalar, in which it equals $R^2(dw/w)^2$. Given two such differentials ("caps") with matching residues on the two sides of a node, identifying $w_Y w_E = u$ produces a smooth curve with a flat cylinder replacing the node: *sewing* with parameter u. The *quadratic-Hodge bundle* $\mathcal{H}$ over the deformation space consists of quadratic differentials with at most double poles and matching residues at the node. A *Jenkins–Strebel* (or Strebel) differential has closed horizontal trajectories off a critical graph, sweeping out flat cylinders; it is *complete* for a multicurve if its cylinder cores are exactly that multicurve.

Lemma 12 (A boundary pencil and its sewings)**.** *Locally at every point of D there are meromorphic quadratic differentials $Q(\tau, s) = \big(q_Y,\ q_E^0(\tau) + s q_D(\tau)\big)$ on $Y \sqcup E_\tau$, depending holomorphically on (τ, s) ($q_E^0(\tau)$ is identified in the proof), with matching nonzero quadratic residue R^2, independent of τ, s, at the node. They belong to the quadratic-Hodge closure of QN. Sewing their double poles in exponential coordinates $w_Y w_E = u$ gives a curve in N, for every sufficiently small $u \neq 0$. The cap q_Y, and hence w_Y, is independent of both τ and s.*

Proof. Work in a labelled finite deformation chart of the central locus. Close the graph of $X \mapsto Q_X N \subset \mathcal{H}_X$, $X \in N$, in the Grassmannian bundle of two-planes in $\mathcal{H}$. For any limiting plane W at $P = (Y, E_\tau)$, its residue kernel (the differentials in W with zero residue at the node) is contained in $\mathbb{C}(0, q_D(\tau))$, by [1, Corollaries 3.11–3.12 and the proof of Theorem 3.9]; by dimension it equals this line.

Uniqueness of the limiting plane (not merely generic finiteness). For $X_n \to P$ realizing W, take the complete one-cylinder Jenkins–Strebel differentials $J_n \in Q_{X_n} N$ given by [1, Corollary 2.7], with core a and circumference one. They are bounded in a local trivialization

of $\mathcal{H}$: otherwise dividing by their divergent norms gives a nonzero limiting differential but horizontal circumference tending to zero; a small natural-coordinate disc at a regular point of that limit contains a uniform smaller disc for the approximants, and a regular point off their critical graphs would lie on a closed horizontal trajectory too short to leave this disc, impossible in a flat coordinate. For any convergent subsequence of J_n, cylinder/residue convergence, as in [1, Lemma 3.10], gives residue $-1/(4\pi^2)$ at the node. Almost every limiting horizontal trajectory is closed: away from the critical trajectories, flows converge for their common period one, and a bounded length trajectory cannot escape down an infinite pole cylinder. Each resulting closed leaf is peripheral to the node, since the approximating leaves have core a. Strebel uniqueness [13, Theorem 2] identifies the limit as $J(P) = (J_Y, J_{E_\tau})$, the pair of one-pole Strebel differentials of perimeter one, with J_Y fixed. Thus every W equals $\mathbb{C}J(P) + \mathbb{C}(0, q_D(\tau))$. The closure of the graph of $\tau \mapsto W(\tau)$ over L (with its reduced structure) is now proper with singleton fibres, hence finite and birational over the smooth curve L, and therefore isomorphic to it. Thus $W(\tau)$ is holomorphic even at formerly exceptional points. A holomorphic fixed-residue section of $W(\tau)$ has Y-part equal to a fixed multiple of J_Y; adding $s(0, q_D)$ gives the asserted pencil.

Sewing an arbitrary vector of the pencil (not only a differential selected by an existence theorem). Approximate the given vector by $(X_n, Q_n) \in Q_{\max}N$. Compact annuli near both limiting double poles are zero-free. On its approximating annulus the nonzero core period $L_n = \oint_a \sqrt{Q_n}$ tends to $2\pi i R$, up to sign, and the coordinate $w_n = \exp\big(\frac{2\pi i}{L_n} \int \sqrt{Q_n}\big)$ is single-valued. Choose its multiplicative constant so it converges to the limiting exponential coordinate on a slightly larger annulus. On a smaller annulus it is then univalent, by uniform convergence and the argument principle, and its image contains a common round subannulus. After rotating Q_n, these round circles are closed horizontal trajectories, so a is a cylinder of Q_n. The two annuli belong to the same maximal a-cylinder: an annulus between homotopic flat cores on a holomorphic quadratic differential contains no zeros, by Gauss–Bonnet. Cutting this cylinder and filling both sides by punctured discs in the displayed coordinates is holomorphic locally, and convergence on compact subsets on both sides gives convergence of these caps to the prescribed pair. The singleton Cylinder Deformation Theorem [11, Theorem 5.1] allows arbitrary shear and positive height for this cylinder without changing the caps, while staying in $\mathcal{A}_N$. For each fixed sufficiently small $u \neq 0$ (uniformly over a relatively compact cap-parameter neighbourhood: common pole-coordinate radii keep the cylinder height positive), choose these deformations to have sewing parameter u; changing coordinate scalars only reparametrizes u. The sewn approximants converge to the sewing of $Q(\tau, s)$, which lies in N by closedness. Integer changes of the sewing marking are harmless because the Dehn twist T_a stabilizes N; the marked lift of the holomorphic sewing family is taken on a simply connected sector of the punctured u-disc, winding around zero giving that twist. No simultaneous realization of intersecting cylinder multicurves is required. □

The next lemma is the heart of the obstruction: sewing the pencil with parameter u yields *three* parameters (τ, u, s) unless the active differential vanishes at the attaching point.

Lemma 13 (Second-order sewing obstruction)**.** *For the pencil in Lemma 12, suppose the unpointed curves E_τ vary. If q_D has either a simple pole or a regular nonzero value at x_τ,*

the map $(\tau, u, s) \mapsto \mathcal{M}_g$ given by the sewn curves has complex rank three at suitable points with arbitrarily small $u \neq 0$.

Proof. Choose a coordinate z at x_τ, holomorphically in τ, for which $q_E^0 = R^2(dz/z)^2$, and write $q_D = (d(\tau)z^{-1} + e(\tau) + O(z))\,dz^2$, $A = sd/R^2$, $B = se/R^2$. The exponential coordinate of $q_E^0 + sq_D$, normalized by $w_E/z \to 1$, satisfies $w_E = z + \frac{A}{2}z^2 + (\frac{A^2}{16} + \frac{B}{4})z^3 + O(z^4)$. For the A-normalized fundamental bidifferential write $\mathcal{B}(w, w') = ((w-w')^{-2} + \beta + O(w, w'))\,dw\,dw'$. A change $w = z + az^2 + bz^3 + \cdots$ changes its constant regular coefficient by $a^2 - b$: substitute $z = w - aw^2 + (2a^2 - b)w^3 + \cdots$ in the singular part; the holomorphic part has unchanged constant coefficient. Consequently $\beta_E(\tau, s) = \beta_E(\tau, 0) + 3s^2 d(\tau)^2/(16R^4) - se(\tau)/(4R^2)$. If $d \neq 0$, its s-derivative is not identically zero; if $d = 0, e \neq 0$, that derivative is everywhere nonzero.

Let η_α, ω_j be normalized holomorphic one-forms on E_τ, Y, and set $e_\alpha = (\eta_\alpha/dw_E)(x_\tau)$, $v_j = (\omega_j/dw_Y)(y)$, $c_* = 2\pi i$. These evaluations are independent of s, because $dw_E/dz(0) = 1$ and the Y-cap is independent of s. Write Ω for the sewn period matrix and Ω_E, Ω_Y for those of its components. The separating-node expansion is

$$\begin{aligned} \Omega_{\alpha\beta} &= (\Omega_E)_{\alpha\beta} + c_* u^2 \beta_Y e_\alpha e_\beta + O(u^3), \\ \Omega_{\alpha j} &= -c_* u e_\alpha v_j + O(u^2), \\ \Omega_{ij} &= (\Omega_Y)_{ij} + c_* u^2 \beta_E v_i v_j + O(u^3). \end{aligned} \tag{1}$$

Here the E-block's displayed coefficients are independent of s, and Ω_Y is constant. (Continuation of ω_j to E starts with $-uv_j\,dw_E/w_E^2$, whose normalized extension has constant regular coefficient $-uv_j\beta_E$; returning to Y gives $u^2 v_j \beta_E\,dw_Y/w_Y^2$, with B_i-period $c_* u^2 \beta_E v_i v_j$. Reversing the components gives the E-block.) The analytic matching construction [4, Theorem 3.3, Section 5.1.1] controls the remainders; their τ, s-derivatives have the same orders, and a u-derivative lowers the order by one.

By Torelli's theorem, some entry $F(\tau) = (\Omega_E)_{\rho\sigma}$ is nonconstant. Choose a general τ with $F'(\tau) \neq 0$, and choose α, j with $e_\alpha v_j \neq 0$, using base-point freeness of the canonical systems. At an s with $\partial_s \beta_E \neq 0$, $\det \partial(\Omega_{\rho\sigma}, \Omega_{\alpha j}, \Omega_{jj})/\partial(\tau, u, s) = -c_*^2 F' e_\alpha v_j^3 (\partial_s \beta_E) u^2 + O(u^3) \neq 0$. The period matrix is holomorphic in the marked curve, so the moduli rank is at least three. □

Theorem 14 (A positive-order zero is necessary)**.** *If a totally geodesic surface has a one-node separating boundary curve with one constant factor, then the active Teichmüller differential vanishes at the attaching point, and the active component has genus $m \geq 2$.*

Proof. First, the unpointed active curve is nonconstant: for $m = 1$, forgetting the sole marking changes no elliptic modulus, and for $m \geq 2$, a constant unpointed projection would put the marked Teichmüller disc in a Bers fibre (a fibre of the forgetful map $\mathcal{T}_{m,1} \to \mathcal{T}_m$, [2, Section 9(B)]). Kra's strict comparison [5, Theorem 3, Section 10] gives, for distinct points in that fibre, ambient Teichmüller distance strictly less than intrinsic Poincaré distance, so Schwarz's lemma for a disc mapping into the fibre contradicts the ambient disc isometry. A Teichmüller differential has finite area, hence at most a simple pole at the sole marking.

Unless it vanishes there, Lemmas 12 and 13 give a rank-three family in the two-dimensional N, a contradiction. The singularity order at the marked point is constant along its generating disc, so it suffices to work at the general points used above. Finally, on an elliptic curve, $H^0(K^2(p)) = H^0(\mathcal{O}(p))$ is one-dimensional, and every nonzero member is a nowhere-zero holomorphic quadratic differential; thus $m = 1$ is impossible. □

Proposition 15 (Arithmeticity of the active curve)**.** *In the configuration of Theorem 14, the square-root orbit of the active differential has field of definition $\mathbb{Q}$; in particular it contains a square-tiled surface.*

Proof. The differential q_D is holomorphic, so forgetting the attachment preserves its Teichmüller disc. Its unpointed image is an algebraic Teichmüller curve (the forgetful map is proper, and the marked zero has finitely many choices). Choose a periodic direction of q_D, with cylinder cores $B = \{b_1, \dots, b_r\}$. Its cusp gives a point of $\overline{N}$ with nodes $a \cup B$; no core is contracted after forgetting the mark, since a holomorphic quadratic differential has neither an inessential flat core nor two homotopic distinct maximal cylinders, by Gauss–Bonnet. Since $\{a\}$ is one class, the dimension bound shows that B is exactly one further N-class.

Proposition 10 gives $K_N = \mathbb{Q}$. On its rational square-root tangent space, integrations over the lifted cylinder cores are rational linear functionals λ_j, and equivalence makes $\lambda_j = c_j \lambda_1$ with $\lambda_1 \neq 0$. Evaluating on a rational tangent vector where $\lambda_1 \neq 0$ gives $c_j \in \mathbb{Q}$ [11, Lemma 7.2]. The finite-area vector $(0, q_D)$ belongs to $\overline{QN}$, and its cylinders persist on approximants in $Q_{\max}N$: the normalized annular coordinates in the proof of Lemma 12 apply to compact subcylinders. By [1, Corollary 2.2], these c_j are the same constants on every approximant with those marked cylinders, depending only on the cylinder curves and N. Cylinder circumferences converge to those of q_D, so its ratios equal these fixed rational numbers (not merely an arbitrary limit of rational ratios).

Take the orientation cover of this *smooth* active differential, or the differential itself if it is a square. In the nonsquare case every downstairs cylinder has two equal-circumference lifts, and these exhaust the upstairs cylinders: all zeros remain zeros, and a cylinder invariant under the orientation involution would contain a fixed regular point. Thus all horizontal circumference ratios upstairs are rational, and [11, Theorem 1.9] gives field $\mathbb{Q}$. The closed square-root orbit has complex dimension two; rational real and imaginary relative periods are dense in its rational period plane, and after scaling one such point has integral periods and is square-tiled. No specialization of a possibly disconnected orientation cover is used. □

The corner argument needs isolation of the two-node cusps of L (points of $\overline{L}$ where one further curve b of E_τ is pinched) in the *whole* two-node stratum intersection $B_S(0)$.

Proposition 16 (Isolation of two-node cusps)**.** *Suppose a cusp of a central one-node boundary curve has exactly two nodes. Then its coarse point p is isolated in the whole intersection $B_S(0)$ with its two-node stratum S.*

Proof. Choose a compatible marked lift $P \in \overline{N}$, with nodes a, b. They are two singleton classes, a being already a singleton along the one-node curve. For $X_n \in N$ tending to P, the complete one-class Strebel differentials $J_{a,n}, J_{b,n}$, each normalized to circumference one,

have canonical limits $J_a(P), J_b(P)$: the bounded-trajectory argument of Lemma 12 applies verbatim. For each, the other node has zero residue, since a nonzero residue would produce an additional cylinder on the approximants. On each normalization component the limit is the Strebel differential with its prescribed positive pole perimeters, other node preimages being retained as punctures, and components with no such pole carry zero; this uses the punctured uniqueness theorem [13, Theorem 2, Remark 3.8]. Their residue vectors are independent, so every limiting plane is $W(P) = \mathbb{C}J_a(P) + \mathbb{C}J_b(P)$.

All two-cylinder caps lie in $W(P)$. For any $r > 0$, let $J_{1,r}(P)$ denote the complete Strebel cap differential with perimeters $1, r$ at the two nodes. We claim $J_{1,r}(P) \in W(P)$. On each fixed X_n, positive integral multitwists $T_a^k T_b^l$ stabilize N, and their horocycle differentials [1, Lemmas 2.4–2.5] are complete two-cylinder Strebel differentials of prescribed modulus ratio k/l. Continuity and density allow all positive ratios, and the projective endpoints are $J_{a,n}, J_{b,n}$: on a fixed smooth X_n, the flat lengths of the fixed curves a, b have positive lower and finite upper bounds on the compact sphere of unit-area differentials, so a cylinder whose modulus ratio tends to zero has height and area tending to zero, its transverse weight vanishes, the limiting horizontal measured foliation is supported on the remaining core, and uniqueness gives the one-class limit. The circumference ratio varies continuously between these endpoints; the first endpoint tends to zero as $n \to \infty$ and the second to infinity, because a small circle about a zero-residue node in $J_a(P)$ or $J_b(P)$ has length tending to zero with its radius, and its approximating circles bound the opposite core length from above. Thus one may choose two-cylinder differentials on X_n with circumferences exactly $1, r$. They are bounded by the same short-trajectory argument, and their limits are $J_{1,r}(P)$ by Strebel uniqueness, proving the claim. Independent cylinder deformations, as in Lemma 12, consequently put *all* sufficiently small sewings of these caps in N.

A second branch would be a translate of N. Suppose another local surface branch over p has a curve in the full two-node stratum. A compatible marked lift N' then has $\{a, b\}$ as one class, with some fixed circumference ratio r, and its complete Strebel limits are the same $J_{1,r}(P)$. Classwise cylinder deformation and passage to the limit give a nonconstant sewing curve $u_a = c_a z^{d_a}$, $u_b = c_b z^{d_b}$ with $c_a c_b \neq 0$, $d_a, d_b > 0$, in this branch; the positive integral exponents come from the rational modulus ratio, and the constants are obtained by a subsequence (in exponential coordinates normalized at the cap critical graphs their moduli are fixed and their phases lie in a compact torus). This curve also belongs to the two-parameter sewing family in N. On a simply connected sector, after a Dehn-twist translate of N', it lifts to a common holomorphic curve in $N \cap N'$, tending to the same marked P.

Here smoothness of the *coarse interior* V_0 is decisive. At every point X of the common curve, the finite stabilizer of X acts transitively on the local branches over the irreducible smooth germ of V_0, so some mapping class fixes X and takes N to N'. There are countably many mapping classes, and their fixed sets on the common parameter curve are analytic; unless one contains the whole curve, their union is countable. Hence one such γ fixes the whole curve and takes N to N'. Continuity gives $\gamma P = P$, so it permutes a, b and must send their two singleton classes for N to two singleton classes for N', a contradiction. No curve of $B_S(0)$ passes through p, proving isolation. □

Corollary 17 (A continued corner without (L))**.** *At such a two-node cusp, (i)–(ii) give a unique local holomorphic section $p(t) \in B_S(t)$, lying in the closures of both incident one-node stratum intersections. This statement does not assert that $p(t)$ is fixed.*

Proof. Use Proposition 16 and Lemma 4. Both partial smoothings occur centrally by Lemma 7. Isolation permits (ii) in a neighbourhood of p, for either one-node stratum. □

5 Curve rigidity and the lifting property (L)

To run the corner mechanism in the one-node case we need to know how the *deformed* boundary curves $C_t \subset \partial V_t$ look; the only tool available is the rigidity of Teichmüller curve maps, which requires maps from curves to moduli spaces with fixed monodromy. The following condition, stated separately and *not* inferred from the hypotheses of the problem, supplies this. (A *fine level cover* of $\mathcal{M}_{h,n}$ is a finite cover that is a fine moduli space, so maps into it have well-defined monodromy.)

(L) *The continued normalization of each boundary curve used below admits a holomorphic family of connected finite algebraic covers with smoothly trivial source, mapping holomorphically to fixed fine level covers of the labelled moduli factors. The images are the given continued boundary curves. On the central fibre these maps are finite orbifold covers of the marked semisimple curve maps.*

Only the curve C_t of Theorem 20 and, when used, its second curve C'_t need satisfy (L); isolated points need no lifting.

Lemma 18 (Pointed curve-map rigidity)**.** *Let $f_t : C_t \to \mathcal{M}_{h,n}$ be a holomorphic family of maps from smoothly trivial finite-area algebraic curves. Assume that the maps are locally liftable to $\mathcal{T}_{h,n}$ and that their mapping-class-group monodromy representations ρ are the same under the given topological identification of the sources (as holds for an actual family of maps to a fixed fine level cover with smoothly trivial source). If f_0 is a Teichmüller-disc quotient map, then the pairs (C_t, f_t) are isomorphic to (C_0, f_0), with the prescribed marking of the source.*

Proof. This is McMullen's argument [6], spelled out to cover the pointed version. Normalize the hyperbolic metrics to curvature -4, so that a Teichmüller disc is isometric for $d_{\mathcal{T}} = \frac{1}{2}\log K$. A hyperbolic element γ of the central source group acts on the target by a pseudo-Anosov mapping class, whose Teichmüller translation length is precisely the length $\ell_0(\gamma)$ of the corresponding source geodesic. Schwarz's lemma for a locally liftable map (the Teichmüller metric is the Kobayashi metric, by Royden's theorem [2, Section 7(A)]) gives $\ell_t(\gamma) \geq \inf_{x \in \mathcal{T}_{h,n}} d(x, \rho(\gamma)x) = \ell_0(\gamma)$; the fixed representation ρ is essential here. The non-domination theorem for marked length spectra of finite-area hyperbolic surfaces, used in [6], now identifies C_t with C_0 as marked Riemann surfaces.

Under this identification lift the maps to the disc and $\mathcal{T}_{h,n}$, with the same equivariance. On the axis of each hyperbolic γ, equality in the inequality forces the lift to map isometrically onto the unique Teichmüller axis of $\rho(\gamma)$; uniqueness and realization of this axis hold for

punctured surfaces as well [9, Theorem 2.1 and its following remark]. Choose two source axes crossing transversely. Their two fixed target axes meet at the central image of the intersection, and at no other point. The lift therefore agrees with the central lift at that point and along either axis; equivariance fixes its orientation along the axis. The identity theorem gives equality of the holomorphic lifts everywhere. □

Lemma 19 (Product form under (L))**.** *For a boundary-curve family satisfying (L), whose central map $f = (f_j)_j$ to the labelled factors is semisimple, the continued maps $F_t = ((F_t)_j)_j$ have the following form after passing to the covers in (L) and identifying each source with the central curve C:*

$$(F_t)_j(z) = \begin{cases} f_j(z), & j \in I, \\ Z_j(t), & j \notin I, \end{cases} \tag{2}$$

where I is the set of initially active coordinates and the Z_j are holomorphic. Every central cusp consequently gives a holomorphic family of stable limits of the same dual graph; all initially active stable factor limits are fixed.

Proof. Work on the connected finite covers supplied by (L). For an initially constant coordinate the map to its fine level space is constant on the central fibre; its monodromy is trivial and remains so by smooth triviality of this actual family of level maps, so on every fibre it lifts to $\mathcal{T}_{h_j,n_j}$. Bers' bounded-domain realization [2, Section 6] shows that this lift is constant: its bounded coordinate functions extend across the finitely many punctures of the algebraic source, and hence are constant on its compactification. Evaluating on a local holomorphic section of the source family shows that $Z_j(t)$ is holomorphic. Apply Lemma 18 to any one active coordinate to identify the marked source with C, and its axis argument to the other active coordinates with the same marked identification; all are then the fixed maps f_j. This proves (2).

Let c be a puncture of the compactification of C. The fixed algebraic maps f_j extend to the proper spaces $\overline{\mathcal{M}}_{h_j,n_j}$, and their values at c are fixed stable pointed curves. The other factors are the smooth curves $Z_j(t)$. Gluing their prescribed marked points according to the original stratum gives the asserted holomorphic stable-curve family in the coarse compactification. Fixed stable factors do not smooth or acquire nodes, and the remaining factors stay smooth, so its graph is constant. No lifted cusp family was assumed. □

6 Conditional completion by coarse corner comparison

Theorem 20. *Under the hypotheses of the problem and the additional boundary-curve property (L), $Y_t = Y_0$ for all t.*

Proof. Work near $t = 0$. By Corollary 3 and the reduction opening Section 4, write $\Delta_\Gamma = \mathcal{M}_{h,1} \times \mathcal{M}_{m,1}$, with first coordinate Y_t and separating node a. Choose a continued curve component C_t of ∂V_t. Theorem 14 gives $m \geq 2$, and (L) with Lemma 19 gives a fixed Teichmüller curve $D \subset \mathcal{M}_{m,1}$ (the image of the disc D of Section 4) such that

$$C_t = \{Y_t\} \times D. \tag{3}$$

Suppose Y_t is nonconstant. A cusp of D has a stable pointed limit R, with nonempty node set B. Clutching gives a holomorphic section

$$p(t) = Y_t \underset{a}{\cup} R \in \overline{C_t}. \tag{4}$$

Its stratum S has nodes $\{a\} \cup B$.

Step 1: $p(0)$ is isolated in $B_S(0)$. For $|B| = 1$, Proposition 16 gives isolation of $p(0)$. For $|B| \geq 2$, if $p(0)$ belonged to a curve component of $B_S(0)$, Lemmas 2 and 4 would put $p(t)$ in the fixed central curve part. This arc is nonconstant, since a fixed stable curve has only finitely many smooth pointed normalization components. In the finite product cover labelling those components, the explicit arc (4) varies only in its Y-coordinate; a branch lies in a central algebraic curve, and the identity theorem makes its other coordinates constant. Its sole active component would be the leaf Y, contrary to Lemma 9. Thus $p(0)$ is isolated in the whole $B_S(0)$, and its continuation $p(t)$ is unique and isolated for small t. If $|B| \geq 2$, the last assertion of Proposition 6 finishes. Hence assume $B = \{b\}$.

Step 2: the second boundary curve. Choose compatible marked lifts of $C_0, p(0)$ in $\overline{N}$. Unpinching the singleton class a, while retaining b, produces a boundary curve C_0' approaching $p(0)$: the marked intersection is smooth and irreducible, contains points approaching a point outside its stratum, and has dimension at most one. Let U be the normalization component obtained by smoothing a. Its coordinate on C_0' is active: a positive power of T_a stabilizes this marked boundary curve and acts on $\mathcal{T}(U)$ as an infinite-order Dehn twist, which cannot fix a point (a remains essential in U, being essential and not isotopic to b on the original surface).

Step 3a: b nonseparating. Then U is the only coordinate of C_0'. Apply (L) and Lemma 19 to its continuation C_t'. The chosen cusp gives the constant section $p(0) \in B_S(t)$. Uniqueness of the continuation (Lemma 4) identifies it with $p(t)$, fixing Y_t by Proposition 6.

Step 3b: b separating. The section has the chain presentation $Y_t \cup_a A \cup_b W$ with $g(A) = r$, $g(W) = s$, where A, W, including their attachments, are fixed (pieces of the fixed R of (4), hence still resting on (L)). Stability gives $r, s \geq 1$: a rational component of the once-pointed one-node R would have at most two special points. If $s = h$, Corollary 8 directly gives $Y_t = W = Y_0$; this uses constancy on the *whole* one-node stratum intersection and requires no choice of end labels. If $s \neq h$, apply (L) and Lemma 19 to C_t'. Its chosen cusp yields $q(t) \in B_S(t)$, $q(0) = p(0)$, whose stable U-coordinate is the fixed curve $Y_0 \cup_a A$. Uniqueness of the continuation gives $q(t) = p(t)$. The genus-h end is intrinsically distinguished, since the end genera differ; its pointed normalization component is respectively Y_0 and Y_t. Thus $Y_t = Y_0$ in this case too. Local constancy extends to the disc by Lemma 1. □

7 Remaining open issues

The unresolved case is a one-node separating stratum with $K_N = \mathbb{Q}$, an arithmetic active curve of genus $m \geq 2$, and an attaching zero of order $k \geq 1$. The canonical limiting plane and fixed cap in Lemma 12 control V_0, not the normalized boundary covers of its

deformation. At two-node cusps, Corollary 17 supplies a continued corner from (i)–(ii) alone, but its normalization components can a priori all move; proving stationarity of that corner, or fixing its active remainder, would close this case. Theorem 20 uses (L) first to fix the active pointed image in (3), and then for the second curve in the nonseparating and unequal-end-genus cases; for cusps with more than two nodes, its isolation argument also depends on (3). Neither (i)–(ii) nor the arithmetic reduction supplies the fixed-monodromy orbifold maps that (L) demands; a source-lifting theorem, or a surface-level replacement, is still needed. An alternative route would produce a fixed nodal unpointed remainder of the active component in every deformed closure and compare its possible attachments; no such persistent remainder has been obtained from (i)–(ii).

Two warnings (not moduli-space counterexamples) show why more control is needed: the normalized compact curves $z^2 = (x^3 - 1)/(1 + tx^2)$ have genus one at $t = 0$ and genus two nearby; and the graphs $y = x^2 + tx^3$ have total projective central equation $Z(YZ - X^2) = 0$ but individual central closure $YZ - X^2 = 0$ (products with $\mathbb{P}^1$ give surfaces with fixed reduced boundary). Lemma 1 supplies boundary holomorphicity, not preservation of finite-cover topology or exclusion of extra central support.

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